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Objectivity of match analysis in football: Testing the level of agreement between coaches' interpretations of video data

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Abstract

Using video data is a widespread procedure in the preparation for an upcoming opponent across all levels of football, but the way coaches interpret this data and use it for player feedback is still not fully understood. Three studies were conducted to investigate the level of agreement between football coaches on tactical questions regarding the opponent when interpreting the same video data. In Study 1 (scouting feed; $N = 15$) and 2 (tactic view feed; $N = 24$), different video viewing angles of the same match were provided to coaches, followed by simple questions regarding the viewed team (e.g., team formation, most striking player in the opening play of the attacking team). Response analyses using descriptive statistics and Fleiss-Kappa statistics showed great diversity regardless of the angle of the feed. Study 3 was a replication study (scouting feed; $N = 16$) using the identical approach as before but used a different match to introduce greater variety of video stimuli. Across all studies there was a high degree of diversity in coach responses and little consensus on basic questions like the adopted formation or the most striking player in the opening play (Fleiss-Kappa coefficients between -.036 [poor agreement] and .236 [fair agreement]). The present research shows that it is problematic to treat information from video feeds as being objective when preparing for the next opponent, as different coaches derive different interpretations from the same data source. Implications for use of video data, and related contributions to coaching research are discussed.

Keywords

Performance analysis, scouting, soccer, tactics

"Before we demand more of our data, we need to demand more of ourselves"[1] (p. 9).

The quote from Nate Silver's bestselling book "The Signal and the Noise" summarizes the idea of the present series of studies: that data is not interpreted uniformly and can lead to very different conclusions and inferences. To empirically test this idea, we choose the context of sport as there is a downright obsession with data and statistics in professional sports.[2] Across all levels of competition, the most commonly used source of data is video data[3] to gain objective information to inform behavior across all levels of competitive football.[4] The present research attempted to test how much agreement there would be[5] amongst different coaches in football who use the same video data to inform inferences in a simulated match analysis scenario.

Within professional football, it has become standard practice to attempt to use data to initiate behavior change in coach, player, and practice behaviors.[6] While the broader field of performance analysis in sports[7,8] has proposed a variety of methods for creating an objective record of performance in various sports contexts by means of systematic observation, professional football has predominantly utilized video feeds to provide feedback to players.[3] This is also reflected in the contemporary job description of a match analyst in football "which largely

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requires them to produce and present 'key' breakdown statistics supported with complementary video compilations of critical game actions and sequences of play"[6] (p. 711). Even though it is common practice to have individuals extract data and draw inferences from video data[5], we are not aware of research on how effective this is in initiating behavior change in football. One reason why the use of video data for match analysis in football might be problematic is that recent evidence suggests that people do not reach similar conclusions when interpreting the same data set, and this is most likely also the case when interpreting video data in football.[4] For example, a research project that has become prominent under the title "Many Analysts, One Dataset"[9] has cast reasonable doubt on the assumption that people reach similar conclusions when interpreting the same data set, as even experts on using the scientific methodology show very little agreement when drawing conclusions from a given data set.

The project by Silberzahn and colleagues[9] should highlight that "data cannot speak for itself". As coaches, game analysts, and officials in sports typically use video data to inform complex inferences and decisions, it seems likely that there would be little agreement on how to interpret the same video data.[10] This also relates to the complexity involved in the development of meaningful feedback to improve performance, something that has been addressed in previous research regarding the delivery of video-feedback.[11,12] In this regard, recent studies have addressed several themes and stakeholders within the performance analysis domain, such as scouts[13] and analysts[4] as well as looking at quantitative parameters,[6] key performance indicators[12,14] and several other predictive and outcome-specific parameters in the performance-related context.[10] While this contributes to our understanding of the performance analytical process, there are still open questions regarding the process of developing information and feedback, such as the need to further understand aspects like coach learning, contextual factors, as well as applied performance analysis practices.[10,15] This is more so the case today than ever before as we witness the development of novel and complex video data being used to analyse football (e.g., special camera systems in the stadium)– all of which keeps growing exponentially.[16] With the complexity of this data being often associated to increased objectivity of the insights drawn from this data, it is imperative to understand whether this is actually the case. This is the context with which the studies in this paper were conducted – to identify the level of agreement and objectivity that is accrued to responses of simple questions regarding specific characteristics of a typical opposition analysis that they encounter regularly. It contributes to our understanding of coach perception of tactics and principles of play, while maintaining a representative design that mimics regular tasks that they undertake. In addition, the field of performance analysis has criticized the lack of research on opposition analysis.[11] Therefore, we conducted three studies to examine diversity of coach responses to qualitative video data as being presented in a typical daily work setting in order to uncover the process of creating meaningful information for athletes to use before a game. While some conditions are modified according to the aims of the respective studies, across all studies, the questions that made the final questionnaire (as well as the response categories derived from the answers) were developed to resemble daily tasks that these coaches are highly attuned to, trained in, and meant to observe regularly. Given that each of the coaches that made the final samples had interacted with such questions very often, their training and domain-specific knowledge was deemed highly adequate to optimally and accurately respond to the questionnaire.

On a more theoretical level, basic research on sensation and perception[17] and judgment and decision making (e.g.,[18,19]) has provided compelling evidence that humans do not perceive the world in the same uniform, objective manner and also do not act to the same visual information in a comparable, uniform way. Instead, human perception and decision-making show a high degree of individual differences and biases that are shaped by situational circumstances and individual dispositions (e.g.,[20]). Pertinent to the present line of investigation, seminal psychological research has shown that humans even miss very salient information in video clips[21] given certain situational or personal circumstances.[22] Based on this empirical evidence and theorizing, we attempted to critically test the common approach of having experienced soccer coaches extract data and draw inferences from video data in football video clips in an attempt of assessing objectivity in match analysis.

The present research

The present research built on and extended earlier work by Franks and Miller[23] who assessed the effects of different types of instructions on novice soccer coaches' (3rd year physical education students) observational accuracy after watching one half of an international soccer match. More specifically coaches were asked to recall technical events during the game (possession, shots, passes, set pieces, crosses, goal-keeper contact). Results of this study showed that different instructions had no significant effect on coaches, but pertinent to the present research, the study showed that PE students only recalled about 40 per cent of the events correctly. The early work by Franks and Miller was extended by Laird and Waters[24] by having experienced, qualified, football coaches (with a minimum of 6 months experience) to recollect accurately the same critical events (possession, shots, passes, set pieces, crosses, goal-keeper contact) during 45 min of a football match. The results of this study show that the probability of 8 qualified coaches with limited experience (youth coaches with the majority only having 1-year experience) recalling critical events accurately was about 60 percent, which led the authors to conclude that coach observation accuracy and recall ability was about 20 percent greater than novice coaches reported by Franks and Miller.[23]

The rationale of the present research was to extend this early eye witness research and tailor it towards a representative opponent analysis scenario with a sample of expert football scouts. Therefore, we were not interested in observation, recall accuracy or statistical concerns like reliability of simple objectively defined events like a shot, pass, or goal-keeper contact, but in more complex information that expert scouts and coaches deem relevant in preparing a team for the next opponent, like "which formation is the team playing" or "which player is particularly striking in the opening play of the attacking team". In this respect, the present research was deliberately tailored to closely resemble the process of deriving information from video observation in an attempt of preparing for an upcoming opponent and thereby making it directly relevant for the opponent analysis process in football. In contrast to previous research,[23,24] we were not interested in recall accuracy of simple, objectively defined situations like a shot or a goal-keeper contact, which is arguably of limited importance in preparing to play a certain opponent, but in the amount of agreement of experienced soccer coaches in identifying more complex information that is deemed important in preparing for an upcoming opponent.

To test for level-of-agreement, we used the Fleiss-Kappa[25,26] methodology (Kappa measure for testing inter-rater reliability of more than 2 raters) to obtain a measure for the strength of agreement between multiple coaches' assessments with a statistical significance test. The inquiry uses one of the most common sources of performance analysis and feedback in football – video data feeds.[3] A collection of video clips was provided to different coaches, who had to use the video information to answer a brief questionnaire assessing different performance indicators and basic questions regarding strategic and tactical behaviour of the team/players presented in the video. Given previous research from different domains, the basic hypothesis that we tested in studies 1–3 was that different coaches would not reach uniform conclusions and interpretations of the same video data and instead would show a lot of heterogeneity in their responses. To test this hypothesis, we conducted a series of studies using two different 20-min video clips filmed from two different angles.

Study 1: Video opponent analysis (Scouting Feed of Stuttgart vs. Mönchengladbach)

Participants

Regarding the sampling criteria and strategy, we followed the guidelines provided by Robinson[27] about participant sampling. 15 football coaches that were at least in the possession of the UEFA B-license participated in the study, with coaching experience ranging from 2 years to 15 years, co-training Bundesliga teams of smaller age groups, as well as head training Regionalliga and Kreisliga teams. Coaches possessing a C-license or lower were excluded from the study. The average age of the participants was 25.1 years ($SD = 4.3$). Before the start of the study, each of the participants was asked to provide consent regarding the data and information that they would provide as part of the study. The study was approved by the local university's ethics board (this also applies to Study 2 and 3).

Stimuli

For the first video stimuli a convenience sample of a random scene was selected from a German Bundesliga game in the season 2020/2021 between VFB Stuttgart and Borussia Mönchengladbach. The scene started in the 10th minute of the game and went on until minute 30:36. Hence, the total duration of the clip was 20.36 min. The video showed the so-called scouting feed and was provided by SPORTCAST GmbH. The scouting feed was chosen because it showed all ten outfield players of both teams at the same time.

Measures

To resemble complex information that expert scouts and coaches deem relevant in preparing a team for the next opponent, we initially asked three soccer experts (all in possession of the UEFA A-coaching-license) to identify the most important information to derive from opponent analyses in preparation for an upcoming game against the opponent. The three experts were asked to give their answers independently and rate the importance of the information for an opponent analysis on a 10-point Likert scale from "not at all important" (1) to "highly important" (10). Based on this we created 5 questions to collect the information that all 3 experts unanimously stated and selected the ones that were rated highest on average in terms of importance for opponent analyses: 1) What tactical system is the team playing?; 2) How would you describe the space occupation (spatial shape) of the team?; 3) How would you describe the opening play of the attacking team?; 4) How would you describe the group tactical behavior of the attacking team?; 5) Which player is particularly striking in the opening play of the attacking team? This procedure also helped clarify the content and concurrent validity of the questionnaire in terms of addressing the right tactical questions. As for construct validity, answers to certain questions such as question 3 and 5 could be analysed for convergence owing to the required congruence in understanding of the concepts in the questionnaire, because these questions relate to the same component of the opening attacking sequence. These were used to measure construct validity, and all the included coach responses showed convergent validity as would be expected in this regard. Finally, the questions were tailored to resemble a lot of the domain-specific themes that surround an opponent tactical analysis, thereby making it highly representative of the daily tasks that coaches undertake.

Procedure

Data collection was conducted online due to contact restrictions during the Covid-19 pandemic via the platform www.soscisurvey.de.

After answering initial questions collecting demographic data, the football coaches watched the 20-min videoclip with the instruction that they would be questioned about the tactical behavior of one team (i.e., VFB Stuttgart). Then the coaches typed in their answers to the 5 open-ended questions outlined in the measure section.

Data analyses

Initially, one of the authors and a research assistant classified answers into categories (inter-rater agreement ranged between 75% and 100% across Study 1–3). The independent coding of the analysts and the subsequent calculation of the inter-rater agreement ([number of agreements / number of categories] *100) was mainly performed to elicit discussions between the analysts regarding the categorization process. For this reason, at any step of the analysis, the two analysts convened to discuss about differences in their answers until an agreement was reached.

Given that all the aforementioned data was based on qualitative ratings, the descriptive analysis of percentages would provide a basic understanding of levels of agreement, alongside performing an inter-rater reliability test with the Fleiss-Kappa[25,26] method which gives a measure for the strength of agreement for nominal and categorical data along with a statistical significance test. According to Landis and Koch[26] Fleiss-Kappa values < than 0.00 show poor levels of agreement, 0.00–0.20 slight agreement, .021–0.40 fair agreement, 0.41–0.60 moderate agreement, 0.61–0.80 substantial agreement, and 0.81–1.00 almost perfect agreement.

Results

In general 93.3 percent of the coaches stated that match analysis was highly relevant for them to prepare for the next opponent.

Question 1. There was enormous diversity regarding the question of what tactical system VFB Stuttgart was playing as can be seen in Figure 1. Some coaches gave multiple responses as they reported different tactical systems when the team was in possession of the ball as opposed to when it was defending. Further, coaches sometimes reported several tactical formations and did not decide on one. Therefore, the 15 coaches gave 30 responses as to the tactical formation of the team that are summarized in Figure 1. According to the Fleiss-Kappa coefficient there was only very slight agreement between the coaches .023 ($p = .328$) and this was not significant.[25,26]

The most striking finding for the first open-ended question was that there was very little agreement as to what tactical formation the team was playing. Only 6 coaches out of 15 agreed that VFB Stuttgart was at some point during the 20 min of the video was playing in a 4-2-3-1 formation, five coaches agreed that they were sometimes playing in a 3-4-3 formation and another three agreed they were playing in a 3-2-4–1 formation. Two coaches respectively identified a

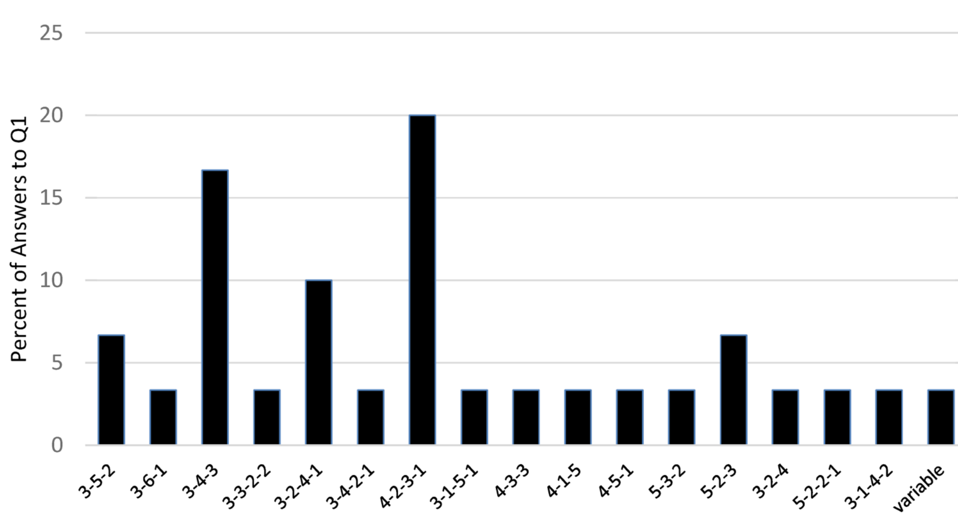

Figure 1. Percent of coaches answers to the question of what tactical formation the team was playing.

3–5-2 and a 5–2-3 formation. 12 other formations were identified by one coach.

Taken together, the results show a clear pattern of very little agreement between coaches when judging the tactical formation of a team based on the identical behavior of players shown in 20 min of video data.

Question 2. Again, there was enormous diversity regarding the question of how coaches would describe the space occupation (spatial shape) of the team. The majority of coaches' responses tended to judge the team shape with generally positive comments with 33 percent of coaches mentioning 'good positioning', 26.7 percent mentioning 'good spatial distribution of players', and 20 percent mentioning 'a high level of flexibility". Interestingly, 26.7 percent of the coaches disagreed and described the positioning and spatial distribution of players as poor and 13.4 percent mentioned that positioning and spatial distribution could be improved. According to the Fleiss-Kappa coefficient there was very poor agreement between the coaches -.036 ($p = .062$).

Altogether 31 different themes emerged in the responses of the coaches. Only on the aforementioned topics did more than two coaches agree. On the other 28 themes there was only agreement between two coaches on 5 themes, while the remaining 23 themes were only mentioned by a single coach.

Question 3. Again, there was enormous diversity regarding the question of how coaches would describe the opening play of the attacking team. There was some agreement between the coaches' answers when neutrally describing some distinctive features of the opening play. 33.3 percent mentioned short-flat passes, 26.7 percent respectively noticed an asymmetric spatial distribution and peculiar positioning, 20 percent mentioned a lack of long-high passes. 13.3 percent (two coaches) agreed on the negative aspects that the team played highly predictively and another 13.3 percent stated that the team showed too little movement. Regarding positive behavior the only theme that there was agreement between two coaches was a high technical quality of the players. According to the Fleiss-Kappa coefficient there was very poor agreement between the coaches -.023 ($p = .131$).

Altogether 39 different themes emerged in the responses of the coaches. Only on the aforementioned topics did more than two coaches agree. The other 28 themes were only mentioned by a single coach with no further agreement.

Question 4. Again, there was enormous diversity regarding the question of how coaches judged the group tactical behavior. There was agreement by two coaches respectively who mentioned good positioning and good technique. Altogether 25 different themes emerged in the responses of the coaches. The remaining 23 themes were only mentioned by a single coach with no further agreement. The Fleiss-Kappa coefficient confirmed this was very poor agreement between the coaches -.032 ($p = .075$).

Question 5. There was also a lot of diversity regarding the question of which player was particularly striking in the opening play of VFB Stuttgart as can be seen in Figure 2. According to the Fleiss-Kappa coefficient there was poor agreement between the coaches -.005 ($p = .839$).

The most striking finding for the 5th question was that 9 different players were mentioned as being particularly important. However, there was a lot of disagreement if the noticed players played a positive or negative role in the opening play of Stuttgart.

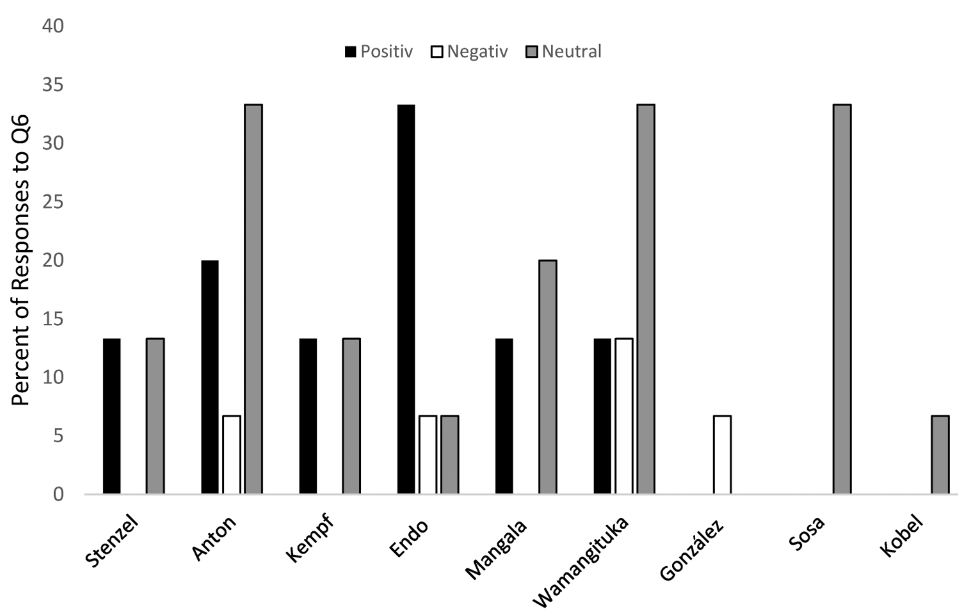

Figure 2. Percent of coaches answers to the question of which player was particularly striking in the opening play of VFB Stuttgart.

Study 2: Video Opponent Analysis (Tactic View of Stuttgart vs. Mönchengladbach)

The aim of Study 2 was two-fold. First, we changed the camera perspective form the scouting feed (from the side of the pitch) to the tactics view (birds eye perspective from behind the goal) to test if this different camera perspective would result in similar amounts of diversity regarding the coach's responses. The rationale for this was to increase the generalizability of the results from Study 1 and exclude the possibility that the pattern of results was mainly due to the specific camera perspective. In addition, we further explored the role of different coaching licenses to the degree of diversity of responses.

Participants

Regarding the sampling criteria and strategy, we followed the guidelines provided by Robinson[27] about participant sampling. 24 football coaches took part in the study. 6 coaches were in possession of the A-license, 8 in possession of the B license, and 10 were in possession of either the C-License, an elite youth coaching license or no license. For the purpose of the present study we distinguished between the high license group (A and B license; n = 14) and the low license group (the rest n = 10). The average age of the participants was 27.7 years (SD = 6.58). Before the start of the study, each of the participants were asked to provide consent regarding the data and information that they would provide as part of the study.

Stimuli and procedure

We used the same sequence of the game between VFB Stuttgart and Borussia Mönchengladbach, only this time it showed the so-called tactical view (birds-eye perspective from behind the goal). Further, we only focused on the first question of naming the tactical formation of the attacking team (Footnote). Otherwise everything was identical to Study 1.

Results

In general 90.3 percent of the coaches stated that match analysis was highly relevant for them to prepare for the next opponent.

There was enormous diversity regarding the question of what tactical system VFB Stuttgart was playing as can be seen in Figure 3. Some coaches gave multiple responses as they reported different tactical systems when the team was in possession of the ball as opposed to when it was defending. Further, coaches sometimes reported several tactical formations and did not decide on one. Therefore, the 24 coaches gave 37 responses as to the tactical formation of the team that are summarized in Figure 2.

Within the high licensed group 38% of the coaches agreed that the team was playing 3-4-3, another 19% agreed on a 3-4-2-1, and another 14% agreed on 3-2-4-1. All the other formations were only mentioned by less than 2 high-licensed coaches. According to the Fleiss-Kappa coefficient there was slight agreement between the high-level coaches .145 ($p = .001$).

The highest level of agreement in the low-licensed group (31%) was on a 3-5-2 formation. 19% of the low-licensed group also identified a 3-4-3 and a 5-2-3 formation. In general, these findings again show a great degree of diversity and limited agreement between football coaches no matter of their license level. According to the Fleiss-Kappa coefficient there was only very slight agreement between the low-level coaches .013 ($p = .788$).

Although, the high licensed coaches showed the highest amount of agreement on a 3-4-3 formation, only 38% of

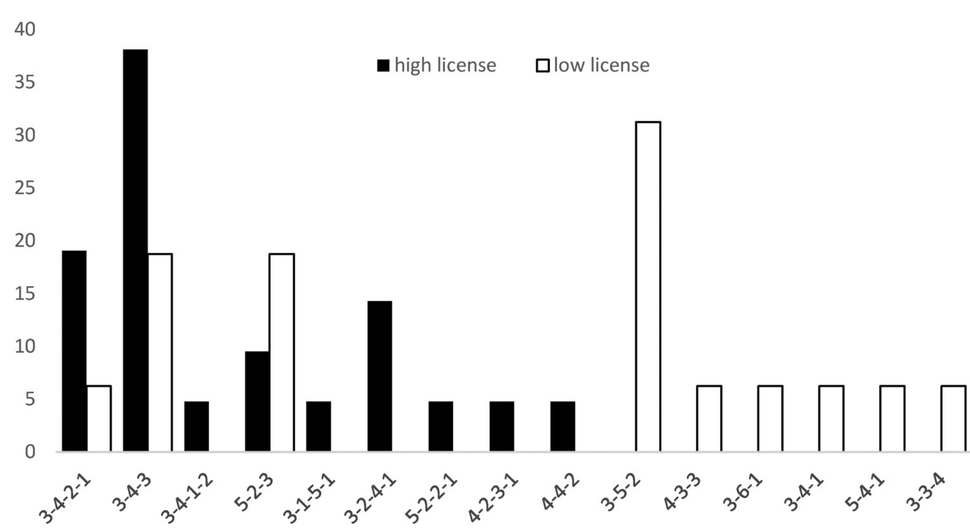

Figure 3. Percent of coaches' answers to the question of what tactical formation the team was playing.

the high-level coaches identified this formation. When comparing the high-level coaches to the lower-level coaches there was slightly more (and statistically significant agreement) amongst the higher-level coaches. However, the amount of agreement is still considered only slight according to statistical convention.

Study 3: Questionnaire responses based on video clips (Fortuna Düsseldorf vs. Mönchengladbach)

Study 3 followed a call by Fiedler,[28] who pointed out the necessity of replicating effects found with one set of stimuli with different stimuli to ensure that the phenomenon of interest does not only apply to a highly specific set of stimulus material, but applies generally to the phenomenon of interest. Study 3 was identical to Study 1 only that we randomly sampled a different video clip for the opponent analysis.

Participants

Regarding the sampling criteria and strategy, we followed the guidelines provided by Robinson[27] about participant sampling. 16 football coaches that were at least in the possession of the UEFA B-license participated in the study, with coaching experience ranging from 2 years to 15 years, coaching teams ranging from co-training Bundesliga teams of smaller age groups, as well as head training Regionalliga and Kreisliga team. Coaches possessing a C-licence or below were not included in this study. The average age of the participants was 31.75 years. Before the start of the study, each of the participants were asked to provide consent regarding the data and information that they would provide as part of the study.

Stimuli

For the second video stimuli another random scene was selected from a German Bundesliga game in the season 2019/2020 between Fortuna Düsseldorf and Borussia Mönchengladbach. The scene started in the 45th minute of the game and went on until the 65th minute lasting for a duration of 20 min. The video showed the so-called scouting feed and was provided by SPORTCAST GmbH. The scouting feed was chosen because it showed all ten outfield players of both teams at the same time.

Procedure

Regarding procedure and data analyses everything was identical to Study 1.

Results

In general 93.8 percent of the coaches stated that match analysis was highly relevant for them to prepare for the next opponent.

Question 1. Although, there was more agreement regarding the question of what tactical system Borussia Mönchengladbach was playing there still was substantial diversity as can be seen in Figure 4. According to the Fleiss-Kappa coefficient there was fair agreement between the coaches .236 ($p = .001$).

Some coaches gave multiple responses. Altogether, the 16 coaches recognized 5 different formations as to the tactical formation of the team with the majority seeing the team play in a 4:3:3 or a 4:2:3:1 formation.

Question 2. As in Study 1, there was enormous diversity regarding the question of how coaches described the space occupation (spatial shape) of the team. The majority of coaches' responses tended to judge the team shape with

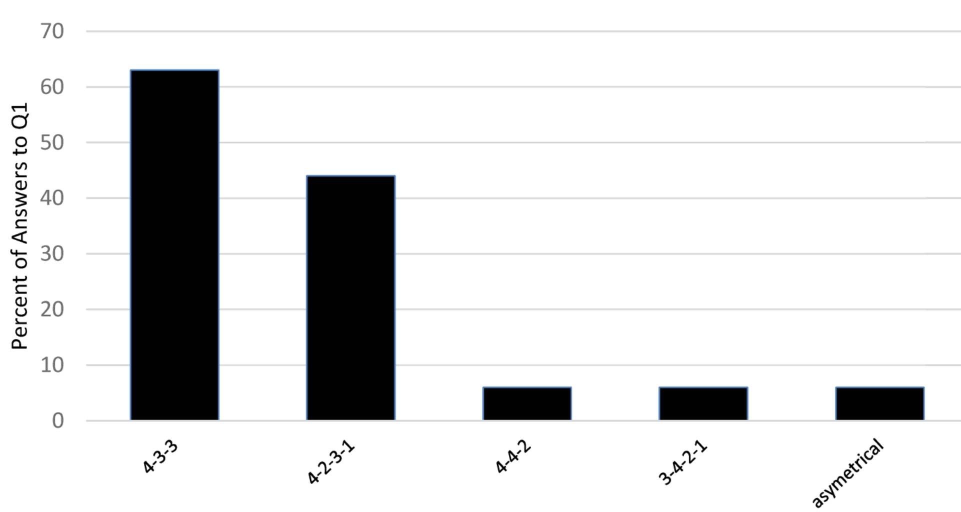

Figure 4. Percent of coaches answers to the question of what tactical formation the team was playing.

generally positive comments with 50 percent of coaches mentioning 'good positioning', 31 percent mentioning 'switching of positions', and 25 percent mentioning 'occupation of free place in the centre". According to the Fleiss-Kappa coefficient there was slight agreement between the coaches .067 ($p = .001$).

Altogether 18 different themes emerged in the responses of the coaches. Only on the aforementioned topics did more than two coaches agree.

Question 3. Again, there was diversity regarding the question of how coaches described the opening play of the attacking team. There was some agreement between the coaches' answers when neutrally describing some distinctive features of the opening play. 31 percent respectively mentioned flexibility of the opening play and passing as distinctive features, 25 percent noticed a conspicuous opening through the center, 19 percent mentioned noticeable shifting to one side, shifting of positions, and that Position 6 (center midfield) often occupied the full-back position. According to the Fleiss-Kappa coefficient there was poor agreement between the coaches -.016 ($p =$ .329).

Altogether 18 different themes emerged in the responses of the coaches. Only on the aforementioned topics did more than three or more coaches agree.

Question 4. There was substantial diversity regarding the question of how coaches described the group tactical behavior. There was agreement by 31 percent of coaches that the group tactical behavior could be regarded as good and 44 percent agreed that the group tactical behavior could be regarded as coordinated. There was also 44 percent agreement that Position 6 (center midfield) often occupied the full-back position and 25 percent agreed on a conspicuous offensive play in which a first deep vertical pass to a striker was bounced back from the striker to a midfield player who again played a direct deep vertical pass. Altogether 22 different themes emerged in the responses of the coaches. Most themes (15) were only mentioned by a single coach with no further agreement. According to the Fleiss-Kappa coefficient there was very slight agreement between the coaches .017 ($p = .360$).

Question 5. There was also a lot of diversity regarding the question of which player was particularly striking in the opening play of Borussia Mönchengladbach as can be seen in Figure 5.

The most striking finding for the 6^{th} question was that 10 different players were mentioned as being particularly important. According to the Fleiss-Kappa coefficient there was slight agreement between the coaches .046 ($p = .139$).

Discussion

The aim of the study was to examine the level of agreement between different football coaches when making inferences based on the same source of video data. Using interpretations of video data is a widespread procedure in the preparation for the next opponent across all levels of football.[3,6] However, we are not aware of any empirical evidence in support of such practice. Therefore, we attempted to initiate the investigation of using video data in an opponent

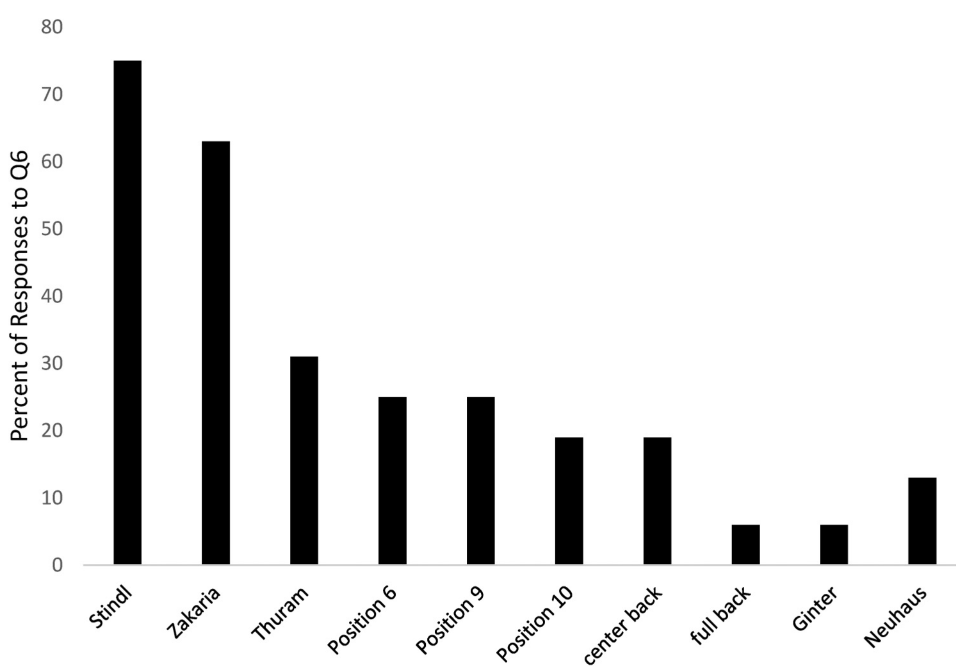

Figure 5. Percent of coaches answers to the question of which player was particularly striking in the opening play of Borussia Mönchengladbach.

analysis scenario by testing the level of agreement of football coaches when answering simple questions based on the same video data. The pattern of results in Studies 1-3 confirm our general hypothesis as responses from coaches show very little consensus in most cases. Fleiss-Kappa coefficients ranged between -.036 and .236 and showed very little agreement in the coaches' responses. The coaches' answers appeared to be highly subjective and involve vast variability of opinions regarding fundamental characteristics of teams such as formation and space control. This is particularly striking as a team's general formation, for example, is openly generalized on live-score websites and other fan-facing services but seems to be highly subjective when it comes to their use in analyses.

Performance analysts are now considered a mainstay of any coaching staff in football, not only at the highest level of sport but also trickling down into academy setups and youth development models. Higher-level clubs now have departments of analysts dedicated at deciphering everything possible about an opponent. While this has taken the sophistication and scope of match analysis to more complex levels, there is still an overwhelming lack of clarity and supporting evidence-base regarding the process of opponent analysis.[3] The present research shows that different coaches see different things in the same video data and derive different conclusions. As all participants in this series of investigation had substantial domain-specific knowledge about football, responses to questions in some themes such as team formation were expected to show some levels of agreement, while responses in other themes were expected to be more diverse. This is seen in the high agreement rate in Study 3 to the question regarding striking player or team formation, which may also point to the effect that broadcast type or perspective of viewing may have on the interpretation of formation and other team characteristics. The hypothesis of expecting diversity was also impacted by the questionnaire consisting of open-ended questions. It must also be noted that the results regarding the thematic diversity of responses is of importance more than the coefficient value of the test statistic. Hence, the present research indicates that the daily practice of using videos in opponent analysis is problematic as it shows to be difficult to derive objective information from video feeds when preparing for the next opponent. Nevertheless, we acknowledge that some caution is warranted when interpreting these results as the participant groups in the present study might not have had the same levels of expertise as some match analysts with year-long experience at professional football clubs. However, results from Study 2 indicated that the amount of diversity in responses was also high in the high-licensed group (which arguably has higher levels of expertise), although the high-licenced group showed slightly more agreement than the low-licensed group when reporting the team-formation.

On a theoretical level, the results support basic research showing high degrees of individuality in perception[17] and judgment and decision making (e.g.,[18,19]). More specifically, the pattern of results supports a general psychological phenomenon: i.e., that people do not appear to interpret data objectively, but rather interpret the data in accordance with current situational and personal factors.[29] When applied to match analysis in sports, it is possible that analysis departments may interpret match analysis data in biased ways.

Applied implications

Coaches and match analysts are under pressure week after week and have to try to prepare their own team as well as possible for the next opponent. To ensure that the data collected from match analyses is helpful and not a hindrance in this endeavor, we consider it essential that future research and applied practice starts to scrutinize on some of the day-to-day work in this field, such as the use of video data in opposition analyses. While recent work has investigated several components of performance analysis such as the adoption of key performance indicators[14,30,31] as well as the subjectivity in the talent identification process,[13] there are still questions that remain unanswered regarding the process by which meaningful information is created for players, specifically while maintaining the structure of daily tasks that are critical in the delivery of applied performance analysis.[4,5,11,12] Furthermore, it also approaches the use of data in football from a domain-specific perspective by asking tactical and descriptive answers to coaches at varying levels of experience. The present research is the first to show that there may be numerous subjective biases when interpreting video data. Consequently, this may also mean that the general belief of greater objectivity accrued to the use of data sources like video feeds may not always be accurate and therefore might not necessarily be helpful in preparing for an upcoming opponent.[10,12,32] A first important step in initiating applied change is to create awareness of a problem as we intended with this research.

Next steps in the efficient use of video feeds in opponent analysis should shed light on the specifics of coach-analyst-player interactions and relationships and how reliable, objective information can be derived and communicated to teams and players.[33] In this respect, process models of decision-making from psychology (e.g.,[34]) can provide reasonable advice on how to proceed with video analyses in football. For example, applied to the context of using video analysis in preparation for an upcoming opponent, these models suggest that 1) coaches are well advised to clearly define their goals and communicate these to other team members in regard to what is expected from match analysis data, what data should be collected, and how these data will be used; 2) It should be clearly specified how the data derived from video analyses can be helpful for achieving these goals. This should help to ensure that only the most important data will be collected and used, and as a side effect might help to improve the quality of collected data; 3) Coaches should critically discuss the practical implications of the video analysis with other people to help make the implications more objective; 4) Finally, the coaching staff should try to reach an agreement in a critical dialogue on what information from the video analysis is communicated to the entire team, what will possibly be communicated to parts of the team or to individual players; 5) During the match, coaches should continuously evaluate whether the communicated information from the video analyses is helpful in reaching the goals and adjust if needed; 6) In the post-match period, match analysts and coaches should evaluate whether the goals have been achieved and if the information derived from video analysis was helpful in this regard. In this way the process of video analysis and data-driven match preparation can be continually optimized and potential problems that arise from subjective interpretations of video data avoided.

Limitations

The present research is not without limitations. First, the present research only focuses on a convenience sample of two video clips from the German Bundesliga and not on a broader sample of soccer games. In addition, the study only focused on video data and did not include other data such as event data or position data to test if inferences drawn from such data sources might be interpreted in a more uniform manner. Further, the study tested for the amount of agreement between different soccer coaches, without testing the individual responses against a "correct criterion" as was done in previous research.[23,24] However, this was done deliberately as the present studies tried to implement a representative design that resembled how coaches and scouts derive information from video clips to prepare for an upcoming opponent. This involved fairly complex inferences like judging "which is the most striking player in the opening play of the attacking team". These complex inferences do not necessarily have a correct answer similar to previous research in which participants were asked to count shots or goal-keeper contacts. Hence, we decided to only test for agreement in responses, instead of accuracy in terms of for example convergence with an experts' response. Finally, despite our attempt of extending prior research with a representative opponent analysis scenario, it remains speculative how results would differ if coaches would actually use video data in preparation for an upcoming opponent as the studies' participants only had to answer 5 questions to a sample video feed in a low stakes situation without actually having to pass on relevant information to their team about an upcoming opponent.

Conclusion

This research does not intend to criticize the use of video data in opponent analysis per se, but attempts to raise awareness of the problems involved. The results from the present study show a substantial degree of subjectivity in interpreting the same video data and are well-aligned with the quote of Nate Silver's best-selling book: "data have no way of speaking for themselves. We speak for them."[1] (p. 9). In this respect, we consider it important to inform coaches and analysist about these findings in assisting them in using video data to usefully inform training protocols and match preparation and not simply use video analysis because everybody else in the field does.

Footnote

Due to space restrictions we only focused on the first question of naming the formation of the teams and do not report the results for Questions 2–5. However, there again were similar degrees of diversity and very low levels of agreement similar to Study 1. Also, there was no pattern that there was more agreement between coaches with a higher coaching license compared to coaches with a lower license.

Open science

All data will be made available at reasonable request to the corresponding author.

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